Priors

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Priors

- Priors: What you believe about parameter before you see the data. Written as probability distribution.
  - Vague/Flat
  - Informative

Informative Priors

- They are based on
  - Statistics reported in literature
  - Posterior distributions from previous studies
  - Meta-analysis
- I was trying to model range expansion by a Neo-native species due to climate change. I ended up using diffused informative prior based on previous studies.
  - Helps speed up model convergence

Vague or Flat Priors

- A vague prior is a distribution with a range of uncertainity that is much larger than range of reasonable values for the parameter.

- Unlike common belief- they influence your posterior but maybe not as much as informative priors.

    - No prior is truly non-informative

How to Choose your prior distribution

- $dx/dt = \alpha x + \beta xy$
  - x and y are the population densities of prey and predator
  - α per-capita growth rate of prey, β is the effect of predator on prey's death
    - β - cannot be positive. Needs to be negative.
      - Maybe $\beta \sim$ uniform (-0.2, 0)
    - α - cannot be negative. Needs to be positive.

Choose your Vague Priors

- Some common examples:
  - For strictly non-negative quantities: Gamma (0.001, 0.001) or log-normal
  - For Variances: Uniform (0, Really large number depending on your data)
  - Regression Coeffecients: Normal (0, variance much larger than mean)
  - Survival/ Detection/Probability: Beta(1,1)

- Your priors are only vague depending on your data.
- α ~ Normal (0,1000)
  - Vague for $\alpha_0 = 2$ (prior almost flat, 0.002SD away from mean)
    - Prior density peak ~ as that of mean
  - But if your $\alpha_0 = 10{,}000$
    - 95% of its probability lies between mean ± 2SD = -2000 to 2000
    - 10,000 is 10 SD away from mean
      - Prior density peak ~ 0
        - This prior says that value of 10,000 is impossible => makes it informative prior
- Prior is steep where the data is